

1.2(a)

$$x_1(t) = [u(t+1) - u(t)] - [u(t) - u(t-1)]$$

$$(c) \quad x_1(t) = \sin[\pi(t-1)] u(t-1)$$

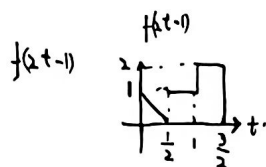
1.10

$$(2) \quad x[4-n]$$



1.7

(1)



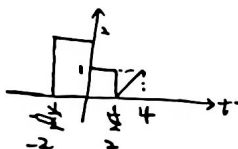
$$(4) \quad x[n-3] \delta[n-3]$$



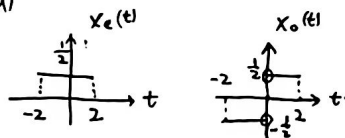
1.12

(3)

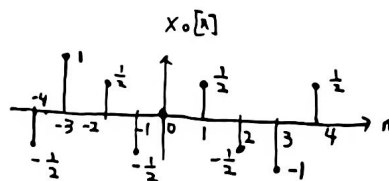
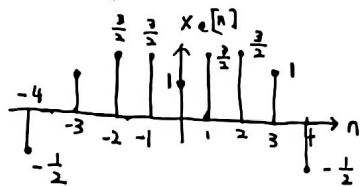
$$f\left(\frac{-t}{2}+1\right)$$



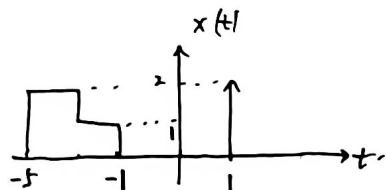
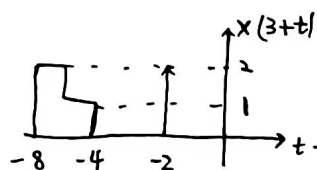
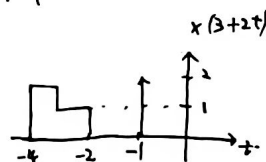
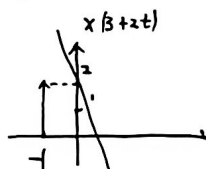
(a)



(b)



1-9



1.14

(2) $a \in \mathbb{R}$ 时

$$T = \frac{2\pi}{a} = \frac{2}{a} \text{ 有周期性}$$

$a \notin \mathbb{R}$ 时无周期性

(4)

无周期性 $m \frac{2\pi}{4} = 8\pi m$ 不可能是整数 ($m \in \mathbb{Z}$)

(6)

$$N_1 = \frac{2\pi}{4\pi} = 8 \quad N_2 = 12 \quad N_3 = 4$$

$N = 24$ 具有周期性, 基波周期是 24.

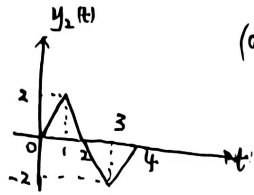
1.16

	记忆	因果	线性	时不变	稳定
(1)	x	✓	x	✓	✓
(2)	✓	✓	x	✓	✗
(3)	x	✓	✓	✓	✓
(4)	✓	x	✓	✓	✓
(5)	x	✓	✓	x	✓
(6)	✓	x	✓	x	✓

1.23

(c) $x_2(t) = x_1(t) - x_1(t-2)$

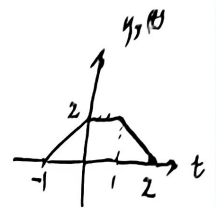
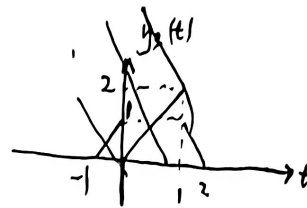
$y_2(t) = y_1(t) - y_1(t-2)$



(d)

$x_3(t) = x_1(t) + x_1(t+1)$

$y_3(t) = y_1(t) + y_1(t+1)$



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$x_2(t) = \frac{dx_1}{dt}$

$y_2(t) = \frac{dy_1}{dt} = \delta(t) - ae^{-at} u(t).$